Gamage Sanjeewa Samrasinghe

MOST OF THE C# TOPICS YOU MAY KNOW AS A DEVELOPER

C# 9



BOOK EDITION NO. 1

Most of the C# Topics You may know as a Developer

Special Thanks for Cotributos of this book

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Most of the C# Topics You may know as a Developer

This is Document of about programming language you must know if it mastering. I believe it must be help full all of C# developers without thinking most of things, directly try to start programming.

Basic Fundamental of Every Programming Language

- Method
- Classes
- Arrays
- Datatypes
- Variables
- Enums
- Switch Statement
- If Statement
- Loops

OOP Principals

- Abstraction
- Encapsulation
- Inheritance
- Polymorphism
- Class
- Objects
- Composition
- Coupling

Data stretcher and Algorithms

SOLID Principals

- Single Responsibility Principle
- Open – Close Principle

- Liskov Substitution Principle
- Interface segregation principle
- Dependency Inversion Principle

Gang of Four Design Patterns 23 -1994

Creational –

This different way to create objects

Structural –

The relational ships between those objects

Behavioral –

The interaction or communication between those objects

Creational – Patterns

1. Abstract Factory
2. Builder
3. Factory Method
4. Prototype
5. Singleton

Structural – Patterns

1. Adapter
2. Bridge
3. Composite
4. Decorator
5. Façade
6. Flyweight
7. Proxy

Behavioral – Patterns

1. Chain of Responsibility
2. Command
3. Interpreter
4. Iterator
5. Mediator
6. Memento

7. Observer
8. State
9. Strategy
10. Template Method
11. Visitor

UML

Advance Topics of C#

- Delegates
- Events
- Generics
- Extension Method
- Lamda Expressions
- LINQ
- Asynchronous (async/await)
- Attribute
- Reflection

Software Development Behaviors

- Software Architecture
- Desktop Application
- MAUI
- Asp.net With MVC Architecture
- Microservices
- JWT Token – Authentication

- Example: - Application Shipping Cart

Data Base Behaviors with Memory management Software Development

- Entity Framework
- Cord Frist
- DB Frist
- Migration
- Table Relationship
- SQLite
- MS SQL
- No SQL Database _MongoDB



ABOUT THE AUTHOR

Gamage Sanjeewa Samrasinghe is an experienced software developer and technical architect specializing in Microsoft technologies and enterprise application development.

KEY TOPICS COVERED (C# 9)

- C# 9 New Features (Records, Init-only Setters, Top-level statements, Pattern matching)
- Object-Oriented Programming
- Language Integrated Query (LINQ)
- Asynchronous Programming (async/await)
- Exception Handling
- Memory Management
- Multithreading & Concurrency
- Generics & Collections
- Delegates, Events & Lambdas
- Attributes & Reflection
- File I/O & Serialization
- ASP.NET Core Fundamentals
- API Development
- Microservices Architecture
- ASP.NET Core Fundamentals
- API Development
- Microservices Architecture Concepts
- Dependency Injection
- Unit Testing with xUnit/NUnit
- Database Access (Entity Framework Core)
- Cloud-Native Development (Azure)
- Design Patterns
- C# Best Practices

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